An IoT System for Real-time Human Activity Recognition

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***Abstract*—This project builds an IoT system for real-time Human Activity Recognition (HAR) that describes human activities with natural language captions. A lightweight model runs locally on the smartphone and immediately produces an activity caption. Selected frames are then uploaded to a server, where a stronger Vision-Language Model (VLM) verifies the local caption and either accepts or rewrites it. However, VLM verification must share an end-to-end latency budget between wireless transmission and server computation, while the VLM execution time varies with the input and the current compute load. We therefore introduce a lightweight latency predictor that estimates the edge-side verification time from the image size, current CPU utilization, and recent VLM execution latency. The remaining end-to-end deadline is converted into a communication budget and used by an application-layer scheduler to prioritize verification uploads from multiple clients.**

I. Motivation

An interactive or embodied system has to respond while the action is still in progress [1]. A closed label such as "sitting" is too coarse for that response [2]. Sitting still, sitting and looking at a phone, and sitting and waving call for different responses, yet they collapse to one class if the recognizer only has a small label set. We therefore describe the activity with an English sentence. Running a strong vision-language model on every captured frame is impractical, because that model is too expensive to call continuously [1]. A small model on the smartphone writes the sentence immediately, and a server with a stronger vision-language model checks it. Sending every picture makes the user wait on the uplink, so the picture is uploaded only when the local sentence is uncertain [3]. Confidence is the minimum probability among the generated tokens. The server then checks the picture and replaces the sentence when the two descriptions disagree.

Checking a selected frame has an end-to-end confirmation deadline. That deadline is shared by wireless transmission and server computation [4]. A fixed communication budget cannot reliably guarantee the deadline, because the checking time varies with the image and the server load. Before a verification frame is uploaded, the system predicts its checking latency and subtracts the predicted computation time from the end-to-end deadline. The remaining time becomes the communication budget of that frame. When multiple clients request verification simultaneously, these budgets determine their upload order.

II. Literature Study

A. Human Activity Recognition

Activity recognizers take RGB, skeleton, IMU, radar, or Wi-Fi, and usually return one label from a closed set. An open-set model can reject an unknown action, but it does not describe it. Free-form captions exist: OV-HAR turns a sensor window into a sentence [2], and OpenHOUSE calls a vision-language model for a description of streaming video [1]. Both are evaluated on recorded data. The missing piece is a sentence produced on the phone, before any server round trip, while the person is still moving.

B. Client–Server Collaboration

Neurosurgeon splits a network by layers, and the user waits for the features [4]. edgeVLM answers on the device and uses a late cloud caption as context for later frames [5]. EdgeBoost uploads an input only at low confidence and still returns a class label [3]. These systems do not show an activity sentence immediately and send the picture only when that sentence is unsafe.

C. Communication

A proportional-fair scheduler and an L4S queue assign service to packets that have already been formed. They do not look at how long the server will spend on the picture after it arrives [6], [7]. Semantic communication scores a transmission by task success, but the payload is still a feature tensor [8]. After the sentence has been written, these mechanisms do not decide whether the picture should follow, how large it should be, and which phone should be checked first. Token confidence decides whether the JPEG is uploaded. JPEG quality decides its size. The predicted server time sets the uplink deadline, and the scheduler serves the earliest deadline first.

III. Technical Approach

Let $\mathcal{C}$ be a bank of English activity sentences, each with a text embedding e_c. For a captured frame x_i, a lightweight generative model on the smartphone writes a short action phrase s_i together with the probability $p_{i,t}$ of each generated token. The phrase is shown at once. Its confidence is $\kappa_i = \min_t p_{i,t}$. The JPEG is uploaded only when $\kappa_i < \tau_{\mathrm{u}}$. Otherwise the phone sends the sentence alone. The server maps that sentence into the bank, $c_{d,i} = \arg\max_{c\in\mathcal{C}} \cos(f_T(s_i), e_c)$. When the JPEG is uploaded, the server also retrieves a caption from the picture, $c_i^\star = \arg\max_{c\in\mathcal{C}} \cos(f_I(x_i), e_c)$. The phone caption stays when $c_i^\star = c_{d,i}$, and is replaced by $c_i^\star$ otherwise. A larger generative model on the server can take the same place as this retrieval checker. It writes its own phrase, the phrase is mapped into $\mathcal{C}$ by the same text encoder, and the caption is replaced when that match differs from $c_{d,i}$.

The on-screen sentence costs only the local time T_i^{local}. A low-confidence frame also waits for the check,

$$T_i^{\mathrm{conf}} = T_i^{\mathrm{wait}} + T_i^{\mathrm{up}} + T_i^{\mathrm{edge}} + T_i^{\mathrm{down}}, \tag{1}$$

* equal contribution.

and meets the deadline when $T_i^{\text{conf}} \le D_i^{\text{conf}}$. Here T_i^{wait} is the time until an upload grant, T_i^{up} is the JPEG transfer, T_i^{edge} is the server check, and T_i^{down} returns a short accept-or-rewrite result.

Before the upload, a small network predicts the check time from the JPEG size L_i, the server CPU utilization U_i^{CPU}, and the previous check time:

$$\hat{T}_i^{\text{edge}} = g_\theta(L_i, U_i^{\text{CPU}}, T_{i-1}^{\text{edge}}). \tag{2}$$

It is trained on traces from the server. Accuracy is the mean absolute error against the measured check time. The uplink budget and deadline are

$$B_i^{\text{up}} = \left[D_i^{\text{conf}} - \hat{T}_i^{\text{edge}} - \hat{T}_i^{\text{down}} - M\right]^+, \quad d_i^{\text{up}} = a_i + B_i^{\text{up}}, \tag{3}$$

where a_i is the arrival time, $\hat{T}_i^{\text{down}}$ is the expected return time, and M is a margin for prediction error. The server grants

$$i^\star = \arg\min_{i \in \mathcal{Q}} d_i^{\text{up}}. \tag{4}$$

A long predicted check leaves less time for the upload, so that request is served sooner. The threshold τ_{u} and the JPEG quality still decide whether a picture is sent and how large it is. The scheduler does not change the Wi-Fi MAC.

IV. Implementation

The client is a smartphone browser. SmolVLM-256M-Instruct [9] runs there in 4-bit form: the vision encoder and the decoder use 4-bit weights, on WASM on an iPhone and on WebGPU where the browser provides it. The sentence and κ_i are ready before any image leaves the phone. Frames below τ_{u} go out as JPEG over Wi-Fi. The server is a laptop running FastAPI. It holds the activity sentence bank and runs CLIP [10] as the checker. Qwen2-VL [11] is the stronger generative checker in the real-world comparison. The phone first sends the JPEG size and the deadline. The laptop reads its CPU utilization and the previous check time, predicts $\hat{T}_i^{\text{edge}}$, and returns an upload grant.

The browser prototype already generates the phrase, shows κ_i, and uploads the JPEG only below the threshold. CLIP replaces the caption when the picture match differs. The predictor, the budget, and the multi-client scheduler are not running yet; measured traces will train the predictor and drive the simulation.

On the phone and the laptop we record local latency, the upload fraction, the accept and rewrite rates, and a human check on a small labeled set. Caption baselines are the phone sentence alone, an upload on every frame, and the confidence gate. The scheduler is simulated with several phones. We report prediction error, deadline satisfaction, the 95th percentile confirmation delay, upload wait, and these figures as the number of clients grows, for round robin, a static budget that uses the average check time, and the predictive budget. Juntao Hu owns the local model, caption bank, acceptance rule, laptop checker, and caption evaluation. Tianchen Liu owns the predictor, the budget, the scheduler, and the multi-client simulation.

TABLE I
Schedule

Dates	Work
23 Sep.–3 Oct.	Local inference and Wi-Fi path
4–17 Oct.	Latency traces and predictor
18–31 Oct.	Dynamic budget and scheduler
1–12 Nov.	Multi-client simulation
13–20 Nov.	Demo, report, and presentation

V. Project Timelines

The smartphone SmolVLM path and the server CLIP check are already running. The remaining work ends before the final presentation on 20 November 2026.

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